

Unit 6: Fluid Statics

Fluid

Substances which flows are known as fluids. Fluids are divided into gases and liquids as it is given in table:6.1

We breathe in fresh air and drink clean water, blood; a lively fluid circulates in human cardiovascular system. Ships to airplanes, automobiles brake systems to balloons, the usage of fluids and their properties is important to technology and everyday life.

Fluid statics, the study of fluid behavior at rest, has many practical applications in various fields such as:

Civil Engineering, Naval Architecture, Petroleum Engineering, Aerospace Engineering and Biomedical Engineering.

Table 6.1 Fluids are differed as:

Property	Gases	Liquids
Definition	Fluids that conform to the shape of their container	Fluids that maintain a fixed volume but conform to the shape of their container
Shape	No fixed shape	No fixed shape
Volume	Conforms to the shape of their container	Maintains a fixed volume
Compressibility	Highly compressible	Slightly compressible
Density	Low density compared to liquids	Higher density compared to gases
Viscosity	Low viscosity	Higher viscosity compared to gases
Surface Tension	No surface tension	Has surface tension
Free Flow	Freely flows	Flows, but not as freely as gases

6.1 Pascal's Law:

The French Philosopher and scientist Blaise Pascal published his treatise on the Equilibrium of Liquids in which he discussed the principles of static fluids. For static fluid the net force on any part of the fluid must be zero, otherwise the fluid will flow.

Pascal's observation - since proven experimentally – provide the foundations for hydraulics, one of the most important developments in modern mechanical technology.

Pascal's Law states that **when a change in pressure is applied at any point to any static fluid, it is transmitted perfectly to all portions of the fluid and to the wall of the container.** For example, as you squeeze your toothpaste tube at one end the same will be transmitted to the opening of tube.

Consider a cylinder of height h and cross-sectional area A with a movable piston of mass m as shown in Fig 6. 1 (a). Adding weight mg at the top of piston increases the pressure at the top, since the extra weight also acts over area of the lid as shown in figure6.1(b). The top of the

piston increases the pressure at the top by $P = Mg/A$ since the additional weight also acts over area A of the lid.

$$P = \frac{F}{A} = \frac{W}{A} \quad \text{--- (6.1)}$$

$$\Delta P_{\text{top}} = \frac{Mg}{A} \quad \text{--- (6.2)}$$

Change in Pressure at upwards =
Change in Pressure downwards

According to Pascal's principle, the

Pressure across the water changes by the same amount i.e. Mg/A . Thus pressure at the downwards also increases by Mg/A .

The pressure at downward of the container is equal to the sum of the atmospheric pressure, the pressure due in fluid and pressure supplied by the mass. The change in pressure at the bottom of the container due to mass is

$$\Delta P_{\text{Bottom}} = \frac{Mg}{A} \quad \text{--- (6.3)}$$

Since the pressure changes are same everywhere in the fluid, we no longer need subscripts to designate the change for top or bottom.

$$\Delta P = \Delta P_{\text{top}} = \Delta P_{\text{bottom}} = \Delta P_{\text{everywhere}} \quad \text{--- (6.4)}$$

Pascal's **barrel** is the name of a hydrostatics experiment allegedly performed by Blaise Pascal, which demonstrated the effects of a changing pressure in a fluid.

Thus, **if an external pressure is applied to a confined fluid, the pressure at every point within the fluid increases by that amount of applied pressure.**

Applications of Pascal's Law:

It has various applications in different fields and It is a fundamental principle with widespread effects in fluid mechanics and has been connected in various fields to enable the transmission of forces and pressures in a controlled and efficient manner.

1. Automobile Hydraulic Brake System:

The rear wheel hydraulic brake system of a front-wheel-drive shown in figure 6.2 is

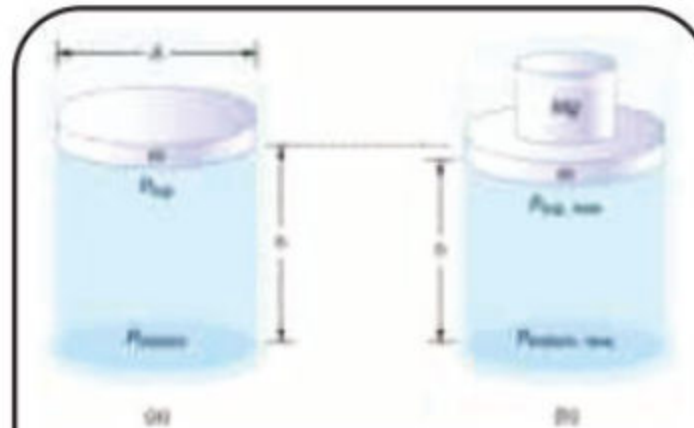


Fig: 6.1

Pressure in a fluid changes when the fluid is compressed.

- (a) Pressure at top of fluid is different from pressure at bottom.
- (b) The increase in pressure by adding weight over the piston, will be same everywhere.

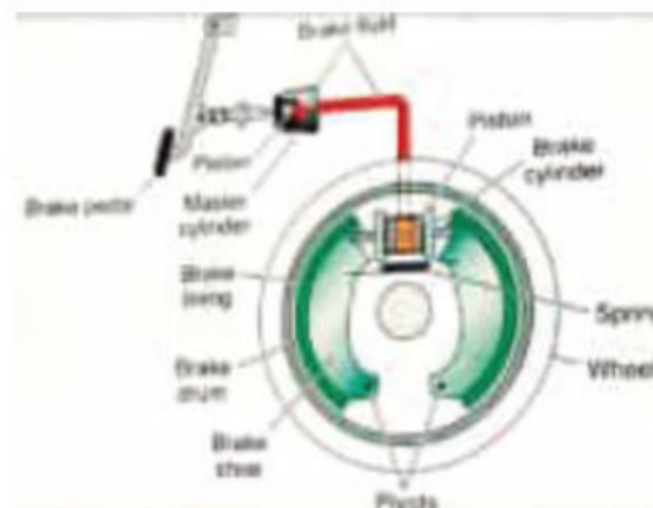


Fig: 6.2

an application of Pascal's principle. When driver pushes the brake pedal, **"The pressure on the piston in master cylinder is transmitted through the brake fluid to the two pistons in the brake cylinder."**

This transmitted pressure then forces the brake-cylinder pistons to push the brake shoes against the brake drum and stop the automobile. Now by releasing the brake pedal releases the pressure on the pistons in the brake cylinder. The spring pulls the brake shoes away from the brake drum, which allows the wheel to turn freely again.

2. Hydraulic Lift or Hydraulic Jack:

Hydraulic lift is able to raise up large weight up to relatively short distance.

The hydraulic lift, or jack is applications of hydraulics being used as a simple machine to multiply force.

It contains an incompressible fluid in a U-shaped tube which is narrower at start and becomes wider area at end which is fixed with a movable piston on each side. If a small force F_1 is applied to the small piston of the hydraulic lift as shown in figure 6.3, the pressure is transmitted with in all directions. The pressure on the large piston is the same as the pressure on the small piston; **however, the force F_2 on the large piston is greater because of its large surface area which used to uplift the car.**

From Pascal's principle, the force needed to lift the automobile is less than the weight of the automobile.

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \dots \dots (6.5)$$

OR

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

Mechanical advantage (M.A) $= \frac{F_2}{F_1}$, which is equal to the ratios of areas.

For example, if the area of output piston is 20 times that of the input cylinder, the force multiplied by a factor of 20. Thus, a force of 200 lbs. could lift a 4000 lbs. of weighing automobile

Activity:

Hold a spoon in a stream of water tap and feel the effect of the differences in pressure



Fig: 6.3

DO YOU KNOW?

When the hand pump handle is pushed down and the piston is raised, air in the pipe is 'thinned' as it expands to fill a larger volume.

Atmospheric pressure on the well surface pushes water up into the pipe, causing water to overflow at the sprout.



Worked Example 6.1

A hydraulic system consists of two connected cylinders, Cylinder A and Cylinder B. Cylinder A has a piston with a radius of 5 cm and Cylinder B has a piston with a radius of 10 cm. A force of 200 N is applied to Cylinder A. Calculate the force exerted by Cylinder B.

Solution:

Data:

Radius of Cylinder A = 5 cm = 0.05 m

Radius of Cylinder B = 10 cm = 0.1 m

Force applied to Cylinder A $\vec{F}_A = 200\text{N}$

Step 1: According to Pascal's law, the pressure exerted on the fluid in Cylinder A is transmitted equally to the fluid in Cylinder B. Therefore, the pressure in both cylinders will be the same.

Step 2: The formula for pressure is:

Pressure (P) = Force (F) / Area (A)

The area of a cylinder can be calculated using the formula:

Area (A) = πr^2

Let's calculate the pressure in Cylinder A:

Area of Cylinder A = πA^2

= $(3.14) (0.05)^2 = 0.00785 \text{ m}^2$

Pressure in Cylinder A $P_A = \vec{F}_A / A$

= $200 \text{ N} / 0.00785 \text{ m}^2$

≈ 25477 Pa

Step 3: Since the pressure is the same in both cylinders, we can calculate the force exerted by Cylinder B using the pressure and the area of Cylinder B:

Area of Cylinder B = πB^2

Area of Cylinder B = $3.14 (0.1\text{m})^2$

= 0.0314 m^2

Step 4: Force exerted by Cylinder B ($\vec{F}_B$) = Pressure (P_A) × Area (B)

= $25477 \text{ Pa} \times 0.0314 \text{ m}^2$

≈ **800 N**

Therefore, the force exerted by Cylinder B is approximately **800 N**, in accordance with Pascal's law.

**Self-Assessment Questions:**

1. How does Pascal's Law relate to the transmission of pressure in a confined fluid?
2. How does the principle of hydraulic systems rely on Pascal's Law?

6.2 Archimedes' Principle:

Archimedes', a Greek physicist of 3rd century, his best-known achievement was his "Eureka" moment when he discovered the principle of flotation. Through which he understood that why objects appear to weigh less when immersed in a denser fluid?

6.2.1 Archimedes' principle:

It states that when an object is immersed into a liquid, it experiences upward thrust which is equal to the weight of the liquid displaced by that object.

The magnitude of the upward force depends upon:

- Volume of the body- more fluid that is displaced, the greater the upthrust.
- Density of fluid- the greater the density, greater is upthrust.

It should be clear from the above that a floating body will displace its own weight of fluid such that there is no vertical resultant force on the body.

Therefore, if a sphere of material density ρ with radius r is fully immersed into a liquid of density σ ; shown in figure 6.4, the apparent weight of sphere is given by

$$\text{Apparent Weight} = \text{Actual Weight} - \text{upthrust} \dots\dots(6.6)$$

The fluid may either be a liquid or air which have low density (about 1 Kg m^{-3}), the upthrust in air is usually small, but sufficient to support a helium filled balloon uplift.

DO YOU KNOW?**Physics in History**

Archimedes' (287 – 212 BC) had been given the task of determining whether a crown made for King Hiero II of Syracuse was of pure gold when he noted the rise in water level while immersing his body in the public baths of Syracuse. Legend reports that he excitedly rushed through streets shouting "Eureka! Eureka!" (I have found it! I have found it!).

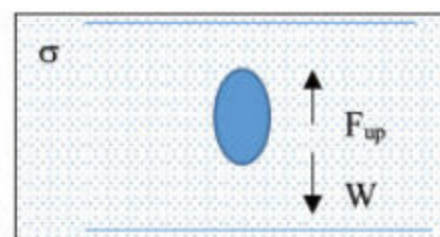


Fig: 6.4

Worked Example 6.2

When a crown of mass 14.7 kg is submerged in water, an accurate scale reads only 13.4 kg . Is the crown made of gold?

Approach:

If the crown is gold, its density and specific gravity must be very high.

Solution:

Step 1: Formula: $W_a = F_r = W - F_B$

$$W - W_a = F_B$$

Step 2: Let V be the volume of completely submerged object and ρ_o the object's density (So $\rho_o V$ is its mass), and let ρ_f is fluid density (water). Then $\rho_f V$ is the weight of fluid displaced, then

$$W = mg = \rho_o Vg$$

$$W - W_a = F_B = \rho_f Vg$$

By dividing them

$$\frac{W}{W - W_a} = \frac{\rho_o Vg}{\rho_f Vg} = \frac{\rho_o}{\rho_f}$$

Step 3: Density of fluid i.e. water is $1 \times 10^3 \text{ kg/m}^3$

$$\frac{\rho_o}{\rho_f} = \frac{W}{W - W_a} = \frac{(14.7 \text{ kg})g}{(14.7 \text{ kg} - 13.4 \text{ kg})g} = 11.3$$

This shows that crown is not made of gold rather made of lead.

Upthrust:

It is the force exerted upwards on an object submerged in a fluid, equal in magnitude to the weight of the fluid displaced by the object. It is a result of Archimedes' principle, which states that the upward buoyant force on an object is equal to the weight of the fluid it displaces. This force helps to support the object and prevent it from sinking. As much as object dips deeper, pressure increases with depth. So, the object has difference in pressure at top and bottom. This difference of pressure produces an upward force which is called upthrust. It is also known as buoyancy or buoyant force.

Upthrust is significantly larger in liquids than in gasses, because liquids are much denser than gasses.

Consider a cylinder of height 'h' and area of cross-section 'A' is immersed in a vessel containing a liquid of density ρ . The upthrust as shown in figure 6.5 is acting on the cylinder from bottom to top is given as

$$\Delta P = \frac{F}{A}$$

$$F_{\text{thrust}} = \Delta P \times A$$

$$F_{\text{thrust}} = -\rho g \Delta h \times A$$

$$F_{\text{thrust}} = -\rho g V \dots\dots(6.7)$$

The equation of upthrust

where

F_{thrust} = Upthrust or Buoyant force

ρ = density of fluid

g = acceleration due to gravity (Upwards)

V = Volume of fluid

If the object has same density as of liquid, then it will stay, neither sinking nor floating just as the liquid nearby banks stays still. But if the object is denser than liquid, then same volume will weigh more and it will sink, because the upthrust will not be enough to support it. If the object has less density, the upthrust will make it float up to surface.

Explore:

Look in the table 6.1 and decide which substances will sink in water. What will afloat on mercury? Will ice float on petrol?

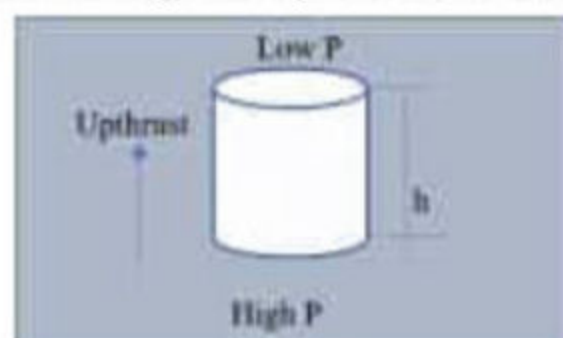


Fig: 6.5

Table 6.1

Substance			Density	
Solid	Liquid	Gas	Kg/m ³	g/m ³
Gold			19000	19
	Mercury		13600	13.6
Iron			8800	8.8
	Water		1000	1
Ice			920	0.92
	Petrol		800	0.8
		Air	1.3	0.013
Lead			11340	11.34

Self-Assessment Questions:

- How is the weight of the displaced fluid related to the upthrust acting on an immersed object?
- How do submarines and hot air balloons take advantage of the principles of Archimedes' theory in their design and operation?

6.3 Buoyancy and law of flotation:

Buoyancy and flotation are fundamental principles governing the behavior of objects in fluids, particularly liquids like water. Buoyancy is the upward force exerted by a fluid on an immersed object, opposing gravity, and enabling objects to float or sink. Flotation is closely related and refers specifically to an object's ability to stay afloat on the liquid surface. Understanding these principles is crucial in various fields, such as shipbuilding, engineering, swimming, and boating, allowing us to comprehend why objects float or sink and design stable structures and vessels in fluid environments.

6.3.1 Buoyancy:

Buoyancy is the upward force exerted by a fluid on an immersed object. It is a result of the pressure difference between the top and bottom surfaces of the object due to the weight of the displaced fluid.

The magnitude of the buoyant force is equal to the weight of the fluid displaced by the object. This can be calculated using Archimedes' principle, which states that the buoyant force is equal to the weight of the fluid displaced by the object.

The formula to calculate the buoyant force is:

Buoyant Force = Density of Fluid × Volume of Displaced Fluid × Acceleration due to Gravity
where:

Density of Fluid is the density of the fluid in which the object is submerged.

Volume of Displaced Fluid is the volume of the fluid displaced by the object.

Acceleration due to Gravity is the acceleration experienced by objects due to the Earth's gravitational field (approximately 9.8 m/s^2).

The buoyant force acts in the opposite direction to gravity, resulting in an apparent reduction in the weight of the object when it is submerged in a fluid.

For an object fully submerged in a fluid, the volume of displaced fluid is equal to the volume of the object itself. If the object is partially submerged, only the submerged portion contributes to the volume of displaced fluid.

The buoyant force is the reason why objects appear to be lighter when submerged in water or other fluids. It is also the principle behind the floating of ships and other vessels, as the buoyant force is greater than the weight of the ship, allowing it to stay afloat.

In liquids:

- If the weight of the submerged object is greater than buoyant force, the object sinks.
- If the weight of object is equal to the buoyant force acting upwards on the submerged object, it remains at any level in fluid, like a fish.
- If the buoyant force is greater than the weight of object which is completely submerged, it rises to the surface and floats.

For example, a ship that is launched into sea, sinks into the ocean until the weight of the water it displaces is just equal to its own weight. As the ship is loaded, it sinks deeper, displacing more

water, and so the magnitude of the buoyant force continuously matches the weight of the ship and its cargo.

In gases:

We live at the bottom of ocean of air and look upward at balloons and other lighter than air objects are drifting above us. Air pressure acting upward against an object immersed in air is greater than the pressure above pushing down. The buoyancy is equal to the weight of fluid displaced. So, Archimedes' principle also implies to air which is stated as:

'An object surrounded by air is buoyed up by a force equal to the weight of the air displaced.'

A cubic meter of air at normal atmospheric pressure and room temperature has a mass of about 1.2 Kg, so its weight becomes around 12 N.

- If the mass of the 1 m^3 object is greater than 1.2 Kg, it falls to the ground, when released.
- If an object of this size has mass of less than 1.2 Kg, buoyant force is greater than weight and it rises in air.

Because gas (Hydrogen or Helium) filled in balloons that rises in air are less dense than air. Air becomes less dense at high altitude; a lesser weight of air is displaced per given volume as the balloon rises. When the weight of displaced air equals the total weight of the balloon, upward motion of balloon ceases. It can be stated as when the buoyant force on the balloon equals its weight, the balloon ceases to rise.

6.3.2 Law of Flotation:

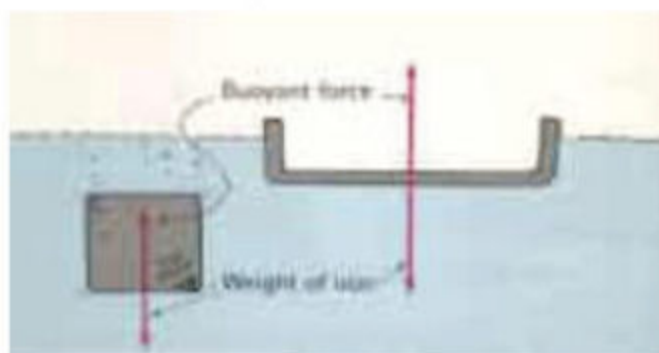


Fig: 6.6

why a ship made of iron floats? Whereas Iron block sink, suppose an iron block is of one ton, so when it is submerged into water, it displaces only $1/8$ ton of water, which is certainly not enough to prevent it from sinking. Contrary, if we reshape same one-ton iron into a bowl shape. When it submerges it into water, it settles into water by displacing a greater volume of water as compared to earlier shape(Block).

DO YOU KNOW?

Floating Mountains

Mountains are bigger than they appear to be. The concept of mountain floating is Isostasy-Archimedes' principle for rocks. Isostasy is the rising or settling of a portion of the Earth's lithosphere that occurs when weight is removed or added in order to maintain equilibrium between buoyancy forces that push the lithosphere upward and gravity forces that pull the lithosphere downwards

Isostasy



DO YOU KNOW?

Iron is nearly eight times denser than water

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The deeper it is immersed, the more water it displaces, and the greater buoyant force acting on it. When the iron boat displaces a weight of water equal to its own weight, it floats. Then principle of flotation is stated as:

'A floating object displaces a weight of fluid equal to its own weight.'

Every ship or submarine must be designed to displace weight of fluid equal to its own weight. Thus a 10K tons ship must be built wide enough to displace 10K tons of water before it is immersed too deep in the water.

In Gases:

A dirigible airship or huge balloon that weighs 100 tons displaces at least 100 tons of air. If it displaces more, it rises. If it displaces less, it descends. If it displaces exactly as its weight, it hovers at constant altitude.

Because the buoyant force upon a body equals the weight of the fluid it displaces, denser fluids exert more buoyant force upon a body than less dense fluids of the same volume. A ship therefore floats higher in salt water and little bit deep in fresh water. In the same way, a solid chunk of iron floats in mercury even though it sinks in water.

Activity:

Try to float an egg in water, it needs dissolve salt in the water until the egg floats. How does the density of an egg compare to that of tap water? To salt water?

DO YOU KNOW?

Scuba diving is an underwater activity where a person breathes from a tank of air while they swim below the surface of the water. The word "scuba" stands for Self-Contained Underwater Breathing Apparatus, which refers to the equipment used to facilitate underwater breathing. This allows a diver to stay underwater for extended periods of time, allowing them to explore the underwater environment, observe marine life, and participate in various underwater activities.

Worked Example 6.3

What volume V of a helium is needed if a balloon is to lift a load of 185 kg?

Step 1: $M = 185 \text{ kg}$

The buoyant force on the helium balloon F_b , which is equal to the weight of displaced air, must be at least equal to the weight of the helium plus the weight of balloon and load, the density of helium is 0.179 kg/m^3 density of air is 1.29 kg/m^3 .

Step 2: $F_b = (m_{\text{hel}} + 185)g$

This equation is written in terms density by using Archimedes' principle

$$\rho_{\text{air}} V g = (\rho_{\text{hel}} V + 185)g$$

Step 3: Solving for V , we get

$$V = \frac{185}{\rho_{\text{air}} - \rho_{\text{hel}}}$$
$$V = \frac{185}{1.29 - 0.179}$$
$$V = 166.5 \text{ m}^3$$



Worked Example 6.4

A solid, square pine wood raft measures 4.5 m on a side and is 0.35 m thick. (a) Determine whether the raft floats in water and (b) if so how much of the raft is beneath the surface the distance h ?

Step 1:

Side = 4.5 m

Thickness = 0.35 m

Step 2:

To determine whether the raft floats, we will compare the weight of the raft to the maximum possible buoyant force and see whether there could be enough buoyant force to balance the weight. If so, then the value of the distance h can be obtained by utilizing the fact that the floating raft is in equilibrium, with the magnitude of the buoyant force equalizing the raft's weight.

Step 3:**(a)**

$W_{\text{raft}} = m_{\text{pine}} g$ where m_{pine} can be calculated by Density $\times$ Volume.

$$W_{\text{raft}} = 550 \times 7.1 \times 9.8$$

$$W_{\text{raft}} = 38269 \text{ N}$$

$$F_B = (\rho V g)_{\text{water}} = 1000 \times 7.1 \times 9.8 = 69580 \text{ N}$$

Since the maximum possible buoyant force exceeds the 38269 N weight of the raft, the raft will float only partially submerged at a distance h beneath the water.

(b)

The buoyant force balances the raft's weight, so $F_B = 38269 \text{ N}$. Using the density of water and volume of water $4 \text{ m} \times 4 \text{ m} \times h$.

$$38269 \text{ N} = W_{\text{water}} = \rho_{\text{water}} \times (4.5 \text{ m} \times 4.5 \text{ m} \times h) \times 9.8$$

Solving for h , we get

$$h = 0.19 \text{ m}$$

Self-Assessment Questions:

1. Why does a steel ship, which is denser than water, float on the surface of the sea according to the principles of buoyancy and the Law of Flotation?
2. Describe the "Law of Flotation" and its application to objects floating in a fluid.

6.4 Surface Tension:

Why liquid drops are spherical, why water rises up in a capillary tube, why certain insects are called pond – skaters who can walk over the surface of water pond. And why the bristles of a paint brush cling together when is taken out of water. All these phenomena can be explained by understanding concept of Surface tension.

The existence and behavior of molecules can be verified by the properties of liquids. Suppose an imaginary free surface of liquid as shown in figure (6.7). The molecules lying on the surface are behaving as tug of war i.e. one side molecules

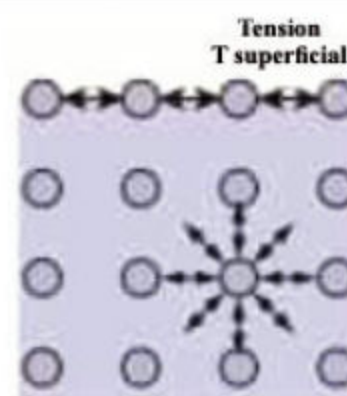


Fig: 6.7 Surface tension

pull to their side while other side molecules to their side in order to decrease the surface area. Hence, surface tension is defined as

“The force per unit length acting on either side of the imaginary line drawn on the liquid surface at rest. The direction of force is tangential to the surface and perpendicular to the line”.

Surface Tension = $\frac{F}{L}$ Its SI unit is N/m or N·m⁻¹.

It is the ability of the surface of a liquid to act like a thin flexible film. As surface of the water to support a needle, The water's surface acts like a thin flexible film

Consider a U-shaped apparatus which encloses a thin film of liquid (Liquid soap film). As shown in figure: 6.8 (a,b) Because of surface tension, a force F is required to pull the movable wire and this increase the surface area of the liquid. The liquid contained by the wire apparatus is thin film having both top and bottom surface. Hence the total length of the surface being increased is $2L$ and the surface tension is

$$T = \frac{F}{2L} \dots (6.8)$$

A delicate apparatus of this type can be used to measure the surface tension of various liquids. The surface tension of water is 0.072 N/m at 20°C. Table shows the standard values of certain liquids

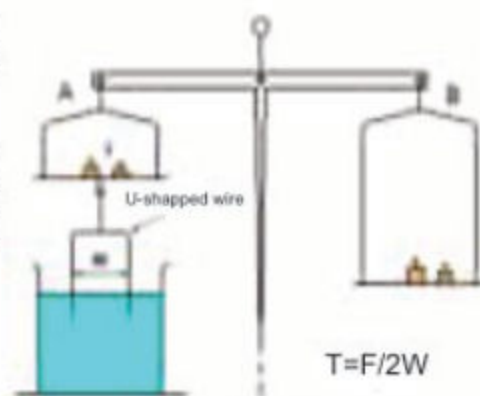


Fig: 6.8 (a)

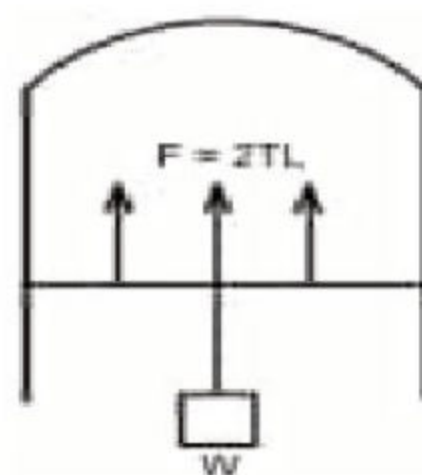


Fig: 6.8 (b)

Table 6.2: Surface Tension of Substances

Substance	Surface Tension N/m
Mercury (20°C)	0.44
Blood Whole (37 °C)	0.058
Blood Plasma (37 °C)	0.073
Alcohol, Ethyl (20°C)	0.023
Water (0 °C)	0.076
Water (20°C)	0.072
Water (100 °C)	0.059
Benzene (20°C)	0.029
Soap Solution (20°C)	0.025
Oxygen (- 193 °C)	0.016

DO YOU KNOW?

The surface tension can be reduced by adding soap or detergent powders called surfactants to laundry water so that the water can more easily be penetrated into the fibers of the clothes being washed.

6.4.1 Examples:

1. When a pin is placed gently over the surface of water, it floats as shown in figure 6.9(a). There is a depression of the surface which behaves like a stretched membrane. As needle is pressed with a finger, the layer of surface tension breaks and the needle sink in the water.

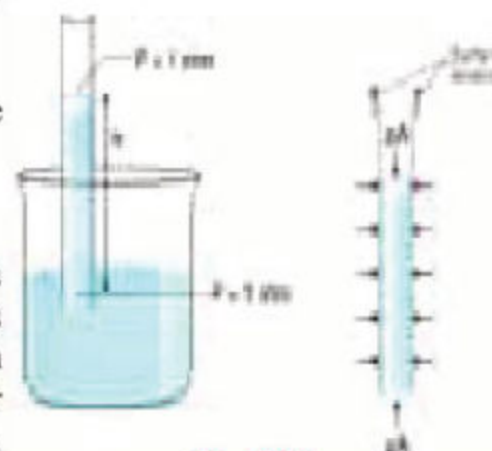
**Fig: 6.9 (a)**

2. Some insects (with long legs) can walk over the water surface. Surface tension of water supports the weight of insect. Actually insect sinks slightly into the fluid so its weight becomes effective weight of that insect, its true weight is less than the buoyant force.
3. A drop of Olive oil is dropped through a pipette, gently inside a mixture of alcohol and water, which forms a perfectly spherical shape. The density of the mixture is same as of olive oil. Due to surface tension, the tendency of a liquid surface is to keep its area minimum. For a given volume, the surface area of a sphere is minimum. Due to this reason, the raindrop is in spherical shape.
4. When a mercury is dropped over glass surface, the globules are spherical in shape.
5. Capillarity
A glass tube of small bore is dipped into water; it rises up the tube a few centimeters. As tube becomes narrower, there is more rise up. Adhesion force between water and glass exceeds cohesion force between water molecules, the meniscus causes up and surface tension.

DO YOU KNOW?

A drop is shown in three different shapes, as

1. Surface tension causes a drop of mercury to be more spherical.
2. Surface tension causes a liquid drop to hold together.
3. The shape of falling rain drop is due to the friction of air.

**Fig: 6.9 (b)****Fig: 6.10**

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This effect is called capillarity or capillary action. The height of column can be calculated by:

$$h = \frac{2T}{\rho r g}$$

T= Surface Tension

r= radius of tube

g= Acceleration due to gravity

ρ =Density of liquid

Worked Example 6.5

The base of an insect's leg is approximately spherical in shape, with a radius of about 2.1×10^{-5} m. The 3.2×10^{-6} kg mass of insect is supported by equally by its six legs. Estimate the angle for an insect on the surface of water. Assume water temperature is 20°C .

Step 1:

$$r = 2.1 \times 10^{-5} \text{ m}$$

$$m = 3.2 \times 10^{-6} \text{ kg}$$

$$\theta = ?$$

Since the insect is in equilibrium, the upward surface tension force is equal to the effective pull of gravity down ward on each leg.

Step 2:

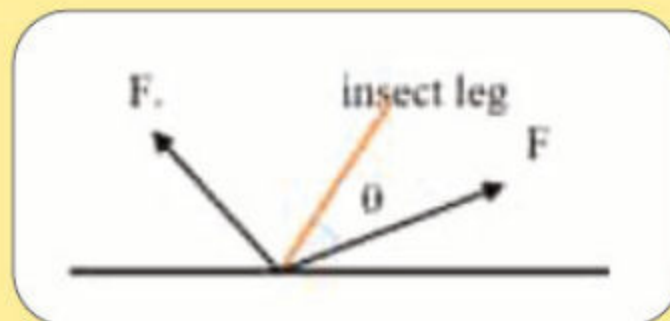
For each leg, we assume the surface tension force acts all around the circle of radius r , at an angle θ . Only the vertical component, $\gamma \cos \theta$, acts to balance the weight mg . So, we set the length L equal to circumference of the circle $L = 2\pi r$. Then the net upward force is due to surface tension is $F_y = (\gamma \cos \theta) L = (\gamma \cos \theta) 2\pi r$. We set this surface tension force γ equal to one-sixth the weight of the insect since it has six legs:

Step 3:

$$2\pi r \gamma \cos \theta = 1/6 mg$$

$$2\pi (2.1 \times 10^{-5}) (0.072) \cos \theta = 1/6 (3.2 \times 10^{-6}) (9.8)$$

$$\theta = 56^\circ$$



Worked Example 6.6

Water flows through a fire hose of diameter 6.5 cm at a speed of 5.99 m/s. Find the flow rate of the fire hose in L/min.

Step 1:

$$v = 5.99 \text{ m/s}$$

$$r = 3.25 \text{ cm} = 0.0325 \text{ m}$$

$$A = \pi r^2 = 3.14 (0.0325)^2$$

flow rate = ?

Step 2:

$$\text{Flow Rate} = Av$$

$$\text{Flow Rate} = (5.99) \times \pi (0.0325)^2$$

$$\text{Flow Rate} = 0.01988 \text{ m}^3/\text{s} \times (1000 \text{ L} / \text{m}^3) (60 \text{ s} / 1 \text{ min})$$

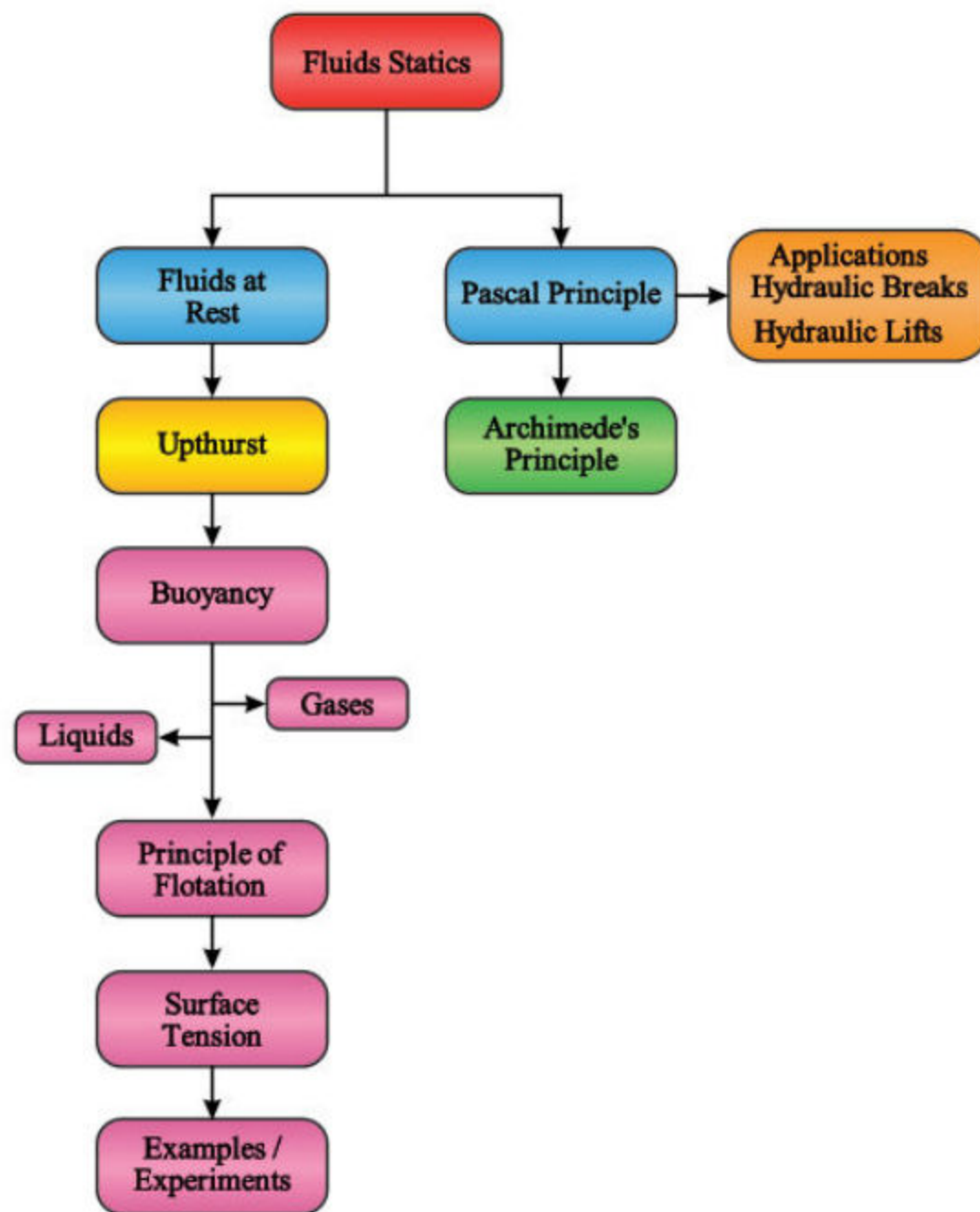
$$\text{Flow Rate} = 1192 \text{ L/min}$$

Self-Assessment Questions:

1. Explain the difference between cohesive forces and adhesive forces in the context of surface tension.
2. How does surface tension impact the behavior of liquid droplets, bubbles, and capillary action?

SUMMARY

- Substances which flows are known as fluids.
- The change in pressure is applied at any point to an enclosed fluid, it is transmitted perfectly to all portions of the fluid and to the wall of the container.
- An object is immersed in a liquid, it experiences upward thrust which is equal to the weight of the liquid displaced by that object.
- An object is submerged partly or completely into a fluid, an upward force is exerted by the fluid. As much as object dips deeper, pressure increases with depth. So, the object has difference in pressure at top and bottom. This difference of pressure produces an upward force which is called upthrust. It is also known as buoyancy or buoyant force.
- Bouncy is the ability of an object to float in fluid.
- A floating object displaces a weight of fluid equal to its own weight.
- The force per unit length acting on either side of the imaginary line drawn on the liquid surface at rest. The direction of force is tangential to the surface and perpendicular to the line.



EXERCISE
Section (A): Multiple Choice Questions (MCQs)

- A completely submerged object always displaces its own _____.
 - weight of fluid
 - volume of a fluid
 - density of a fluid
 - Area of fluid
- The pressure exerted on the ground by a man is greatest when:
 - he stands with both feet flat on the ground
 - he stands flat on one foot
 - he stands on the toes of one foot
 - he lies down on the ground
- In a stationary homogeneous liquid:
 - pressure is the same at all points
 - pressure depends on the direction
 - pressure is independent of any atmospheric pressure on the upper surface of the liquid
 - pressure is the same at all points at the same level
- One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Δp the pressure at the larger piston:
 - increases by $2\Delta p$
 - increases by $\Delta p/2$
 - increases by Δp
 - increases by $4\Delta p$
- In a vacuum, an object has
 - No buoyant force
 - no mass
 - no weight
 - none of these
- The pressure at the bottom of a pond does NOT depend on
 - Water density
 - the depth of the pond
 - the surface area of the pond
 - None of these
- A rock suspended by a weighing scale weighs 5N out of water and 3 N when submerged in water. What is the buoyant force on the rock?
 - 3 N
 - 5 N
 - 8N
 - 15 N
- "An object completely submerged in a fluid displaces its own volume of fluid". This is:
 - Pascal's paradox
 - Archimedes' principle
 - Pascal's principle
 - true, but none of the above
- Salt water has greater density than freshwater. A boat floats in both freshwater and saltwater. The buoyant force on the boat in salt water is ----- that in freshwater.
 - equal to
 - smaller than
 - larger than
 - same as
- You fill a tall glass with ice and then add water to the level of the glass's rim, so some fraction of the ice floats above rim. When the ice melts, what's happens to the water level?
 - the water overflows the rim
 - the water level drops below the rim
 - the water level stays at the rim
 - it depends on the difference in density between water and ice

SRQ's

1. State Pascal's principle. Describe its two applications.
2. (a) Why must a liquid and not a gas be used as the 'fluid' in a hydraulic machine?
(b) On what other important property of a liquid do hydraulic machines depend?
3. Why don't ships made of iron sink?
4. Why do you float higher in salt water than in fresh water?
5. What is the difference between being immersed and being submerged in water?
6. Define surface tension and give its any two applications.
7. A swimmer dives off a raft in a pool. Does the raft rise or sink in the water? What happens to the water level in the pool? Give reasons for your answer.
8. Distinguish between flotation and upthrust.

ERQ's

1. Discuss buoyancy in liquids and gases.
2. Describe law of flotation in liquids and gases.
3. Discuss surface tension with at least three experiments.
4. Explain Archimedes principle and find gold purity by using density.

Numericals

1. In a hydraulic press a force of 20 N is applied to a piston of area 0.20 m^2 . The area of the other piston is 2.0 m^2 . What is (a) the pressure transmitted through the fluid; (b) the force on the piston?
(200 N, 100 Pa)
2. The pressure in a water pipe in the ground floor of a building is $4 \times 10^5 \text{ Pa}$ but three floors up it is only $2 \times 10^5 \text{ Pa}$. What is the height between the ground floor and the third floor? The water in the pipe may be assumed to be stationary; density of water $= 1 \times 10^3 \text{ Kg/m}^3$
(20 m)
3. The small piston of hydraulic press has an area of 10.0 cm^2 . If the applied force is 50.0 N, what must the area of the large piston to exert a pressing force of 4800N? **(960 cm^2)**
4. Mechanical advantage of a hydraulic jack is 420. Find the weight of the heaviest automobile that can be lifted by an applied force of 55 N.
(23000 N)
5. A flat -bottom river barge is 30 ft wide, 85 ft long and 15 ft. deep, (a) how many m^3 of water will displace while the top stays 1 m above the water.? (b) What load in tons will the barge contain under these conditions if the empty barge weighs 160 tons in dry dock?
($3.1 \times 10^4 \text{ ft}^3$, 800 tons)
6. A canal lock gate is 20 m wide and 10 m deep. Calculate the thrust acting on it assuming that the water in the canal is in level with the top of the gate. Density of water is 1000 kg/m^3
($9.8 \times 10^4 \text{ N}$)

7. A tank 4 m long, 3 m wide and 2 m deep is filled to the brim with paraffin (density 800 kg/m^3). Calculate the pressure on the base? What is the thrust on the base?
(16000 Pa, 19200 N)
8. A rectangular boat is 4.0 m wide, 8.0 m long, and 3.0 m deep. (a) How much water will it displace if the top stays 1.0 m above the water? (b) What load will the boat contain under these conditions if the empty boat weighs $8.60 \times 10^4 \text{ N}$ in dry dock?
(Weight density of water = 9800 N/m^3) **(64.0 m³, $5.41 \times 10^5 \text{ N}$)**
9. A hot air balloon has a volume of 2200 m^3 . The density of air at temperature of 20°C is 1.205 kg/m^3 . The density of the hot air inside the balloon at a temperature of 100°C is 0.946 kg/m^3 . How much weight can the hot air can lift?
(5590 N)
10. A spherical balloon has a radius of 7.15 m and is filled with helium. How large a cargo can it lift, assuming that the skin and structure of the balloon have a mass of 930 kg? Neglect the buoyant force on the cargo volume itself.
(920 kg, $9 \times 10^3 \text{ N}$)

Unit

7

Fluid Dynamics

Teaching Periods

9

Weightage %

6



Wind energy form at Gharo: Energy driven through air.

In this unit student should be able to:

- Describe that real fluids are viscous fluids.
- Describe that viscous forces in a fluid cause a retarding force on an object moving through it.
- Explain how does the magnitude of the viscous force on an object moving in fluid depend on the size and velocity of the object.
- Apply Stokes' law to derive an expression for the terminal velocity of spherical body falling through a viscous fluid.
- Use the equation of terminal velocity to solve problems.
- Define the terms: steady (streamline or laminar) flow, incompressible flow and non-viscous flow as applied to the motion of an ideal fluid.
- Explain that at a sufficiently high velocity, the flow of viscous fluid undergoes a transition from laminar to turbulence conditions.
- Describe that the majority of practical examples of a fluid flow and resistance to motion in fluids involve turbulent rather than laminar conditions.
- Identify that the equation of continuity is a form of the principle of conservation of mass.
- Solve problems by using the equation of continuity.
- Describe that the pressure difference can arise from different rates of flow of a fluid (Bernoulli's Effect).
- Interpret and apply Bernoulli Effect in the: filter pump, Venturi meter, atomizers, flow of air over an aero foil and in blood physics.

Fluid Dynamics:

In our daily life, we observe the motion of fluids i.e. gases and liquids through pipes, ducts and passage ways. Examples in our daily life are – water streams from a fire hose, blood courses through our veins. Why does rising smoke curl and twist? How does a nozzle increase the speed of water emerging from nose? How does the body regulate blood flow? The physics of fluids in motion -fluid dynamics-allows us to answer these and many other queries.

Aircraft move through the air, when a fluid is in motion or an object move through a fluid, the pressure within fluid varies with velocity. The forces generated by this pressure difference were explained in 1852 by the German physicist Gustav Magnus. He resolved the problems of, why projectiles spinning about an axis other than their direction of motion will curve off course. Such spinning is played technically by bowler through bowling as swing of ball. Air dynamics is used in throwing and kicking the ball in cricket and football respectively. These phenomenon plays vital role science and technology to make our games exciting.

7.1 Fluid Friction:

7.1.1 Real Fluids are viscous fluids:

Viscous fluids are fluids that resist deformation and flow. They have a high resistance to shear or flow and exhibit internal friction between adjacent layers of the fluid. This internal friction is responsible for the viscosity of the fluid.

Viscous fluids can be found in many aspects of our daily lives. Here are a few examples:

Honey or Syrup: Honey and syrups are examples of highly viscous fluids. When you pour honey or syrup from a container, it flows slowly and tends to stick to the spoon or container. The resistance to flow and the slow pouring rate are due to the high viscosity of these fluids.

Motor Oil: Motor oil used in engines also exhibits viscosity. It is designed to have a certain viscosity to provide lubrication and minimize friction between engine parts. High-viscosity motor oil is used in engines with larger clearances, while low-viscosity oil is used in engines with smaller clearances.

Paint or Varnish: Paints and varnishes also exhibit viscosity. When you apply paint to a surface, it adheres and spreads according to its viscosity. High-viscosity paints are thicker and tend to form thicker layers, while low-viscosity paints flow more easily and may result in thinner coatings.

These examples highlight the behavior of viscous fluids in different situations. Viscosity plays a crucial role in various aspects of our daily lives, from cooking to transportation to manufacturing processes. Understanding and controlling the viscosity of fluids is essential in numerous industries and scientific fields.

7.1.2 Viscous force in a fluid:

Viscous force is an opposition between layers of a fluid. Viscous forces in a fluid are proportional to the rate of change of velocity of fluid's layers. The viscosity of a fluid serves as the proportionality constant.

Suppose a viscous fluid between two plates in which one plate is stationary and other is moveable as shown in (figure 7.1).

The fluid is directly in contact with each plate is held to surface by the adhesive force between the molecules of the liquid and those of the plates. Thus, the upper surface of the fluid moves with the same speed v as the upper plate, whereas the fluid in contact with the stationary plate remains stationary.

The stationary layer of fluid retards the flow of the layer just above it, which in turn retards the flow of next layer, and so on. Thus, the velocity varies continuously from 0 to v through the fluid. The increase in velocity divided by the distance over which change is made – equal to $\frac{v}{l}$ is called the **velocity gradient** which is defined as '*the rate of change of velocity with distance normal to the direction of flow of the layers of the fluid with respect to object passing through the fluid*'.

Viscosity is a fluid's resistance to flow. Fluids resist the relative motion of the immersed objects through them as well as to the motion of layers with differing velocities within them.

For a given fluid, it is required force F which is proportional to the area of fluid in contact with each plate ' A ', and to the speed, ' v ', and is inversely proportional to the separation ' l ', between the plates as shown in figure 7.1.

$$F \propto v \frac{A}{l}$$

For more viscous fluid, the greater force is the required.

Hence, the proportionality constant for this equation is defined

as the coefficient of viscosity η which is intrinsic property of the fluid.

$$F = \eta A \frac{v}{l} \text{ ----- (7.1)}$$

$$\eta = \frac{F/A}{v/l}$$

The SI unit for η is $\text{N}\cdot\text{s}/\text{m}^2 = \text{Pa}\cdot\text{s}$ (Pascal. Second). In CGS system, its unit is called dyne. s/cm^2 , which is called a Poise (P). The temperature has a strong effect; the viscosity of liquids such as motor oil decreases rapidly as temperature increases, while for gases it increases.

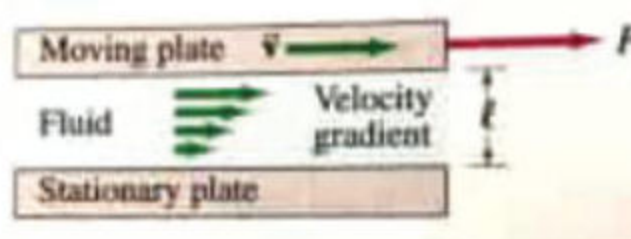


Fig: 7.1
viscous fluid between two plates

Table 7.1: Coefficients of Viscosity

Fluid (Temperature in $^{\circ}\text{C}$)	Coefficient of viscosity, η (Pa. s)
Water (0°)	1.8×10^{-3}
(20°)	1.0×10^{-3}
(100°)	0.3×10^{-3}
Whole Blood (37°)	$\approx 4.0 \times 10^{-3}$
Blood Plasma (37°)	$\approx 1.5 \times 10^{-3}$
Engine Oil (30°)	2.0×10^{-1}
Glycerin (20°)	1.5×10^{-1}
Air (20°)	1.8×10^{-5}
Hydrogen (0°)	9.0×10^{-6}
Water Vapor (100°)	1.3×10^{-2}

Self-Assessment Questions:

1. What is viscosity, and how does it relate to fluid friction?
2. Describe the relationship between fluid velocity and fluid friction.

7.2.1 Derive an expression for terminal velocity of spherical body:

Stokes' law describes the drag force experienced by a small spherical object moving through a viscous fluid. The drag force acting on the object is directly proportional to its velocity and the viscosity of the fluid, and it is given by the equation:

$$\text{Drag Force} = 6\pi\eta r v \dots \dots (7.2)$$

where:

Drag Force is the force experienced by the object due to the fluid drag (measured in newtons, N).

η (eta) is the dynamic viscosity of the fluid (measured in Pascal-seconds, Pa·s or N·s/m²).

r is the radius of the spherical object (measured in meters, m).

v is the velocity of the object relative to the fluid (measured in meters per second, m/s).

The terminal velocity of a spherical body occurs when the drag force equals the gravitational force acting on the object. At terminal velocity, the net force on the body is zero, resulting in constant velocity.

Considering the gravitational force given by:

$$\text{Gravitational Force} = (4/3)\pi r^3 \rho g$$

where:

ρ (rho) is the density of the spherical body (measured in kilograms per cubic meter, kg/m³).

g is the acceleration due to gravity (measured in meters per second squared, m/s²).

At terminal velocity, the drag force is equal to the gravitational force, so we can equate the two expressions:

$$6\pi\eta r v = (4/3)\pi r^3 \rho g$$

Simplifying the equation:

$$6\eta v = (4/3)r^2 \rho g$$

Dividing both sides by 6η :

$$v = (2/9)(r^2 \rho g) / \eta \dots \dots (7.3)$$

This equation gives us the expression for the terminal velocity of a small spherical body moving through a viscous fluid according to Stokes' law. The terminal velocity is directly proportional to the square of the radius of the body (r^2), the density of the body (ρ), and the acceleration due to gravity (g), and inversely proportional to the dynamic viscosity of the fluid (η).

Self-Assessment Questions:

1. What factors affect terminal velocity?
2. Does the mass of an object affect its terminal velocity?